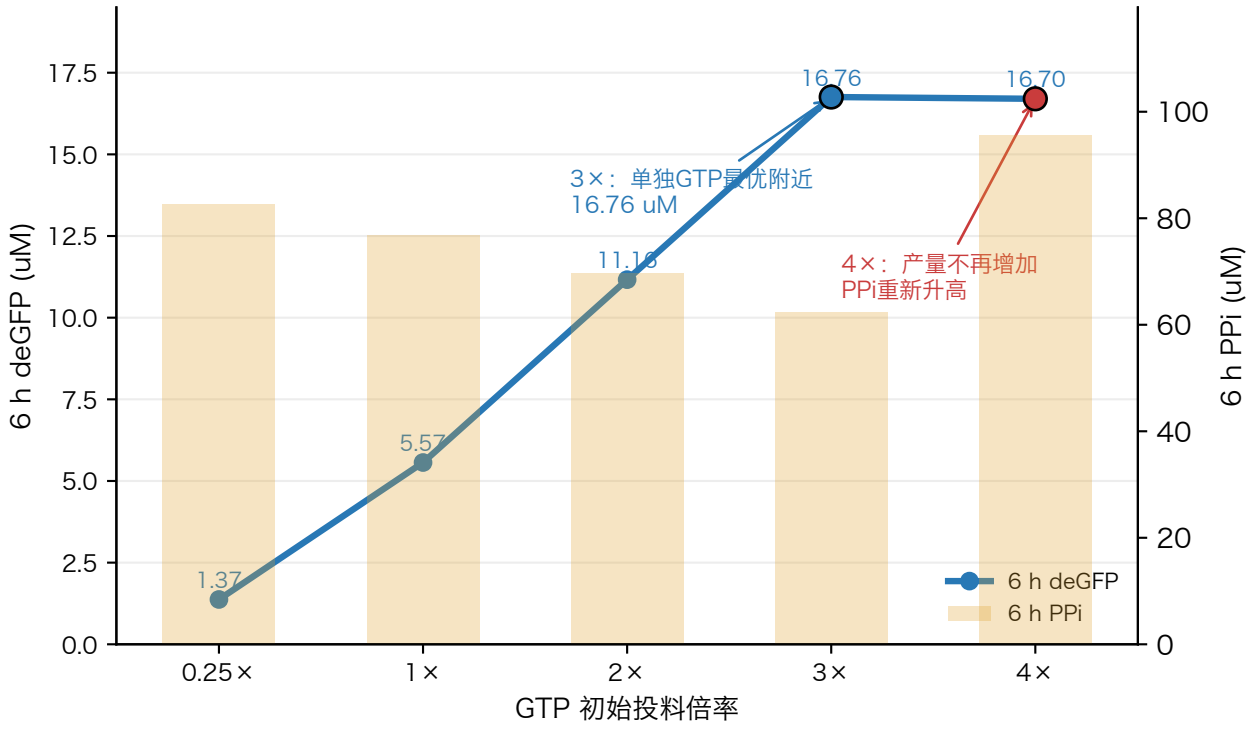
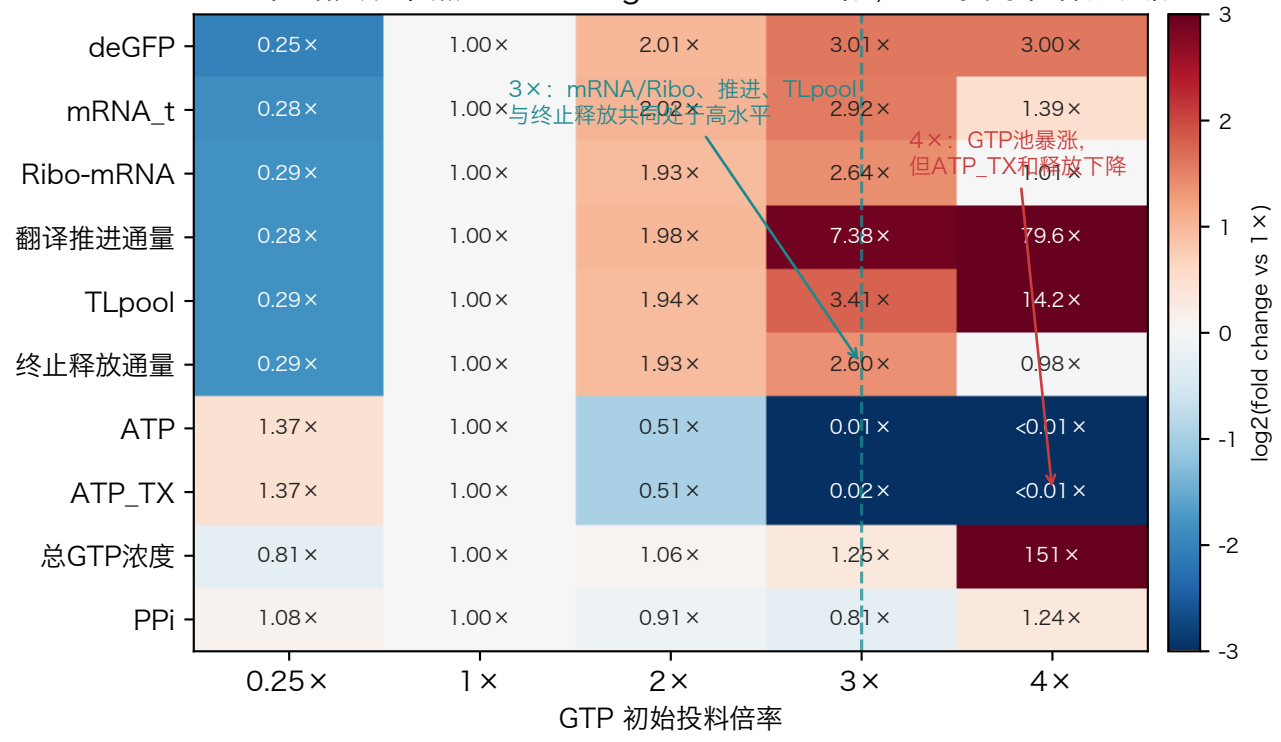
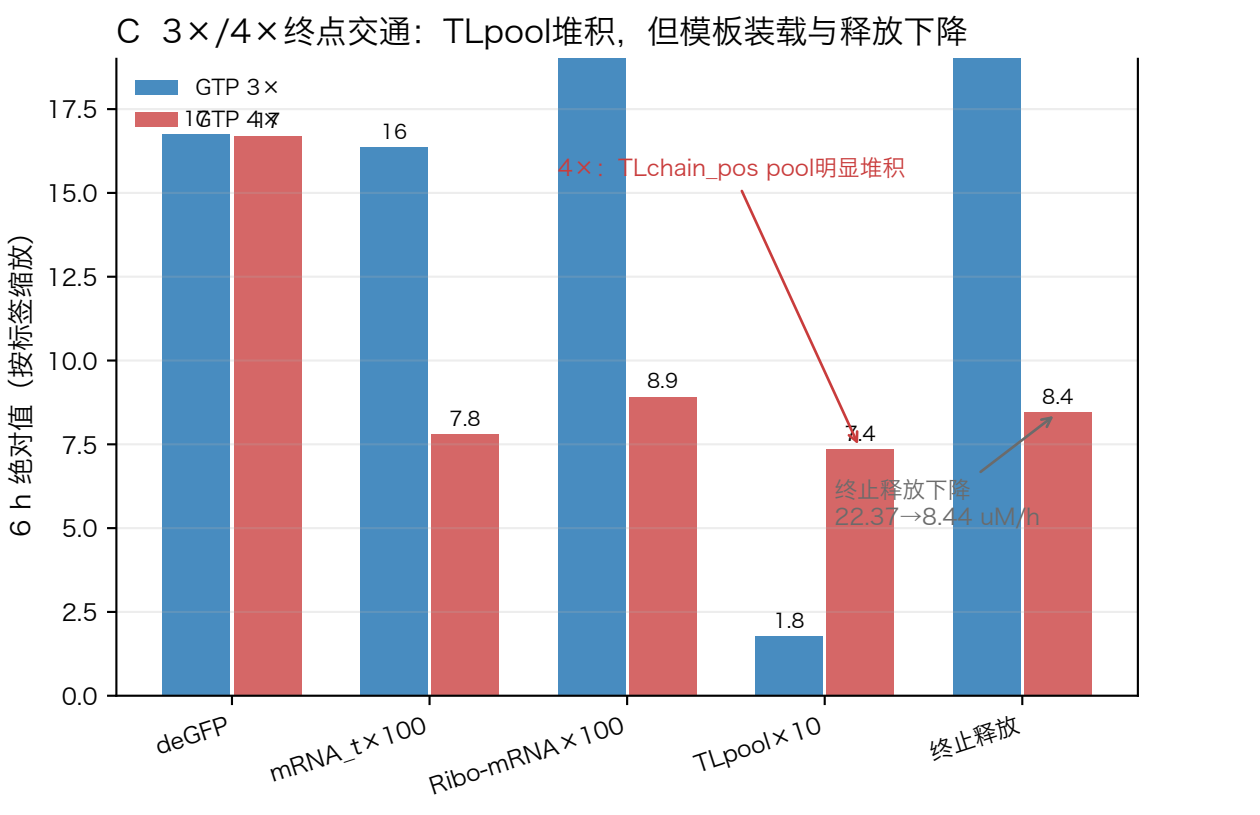
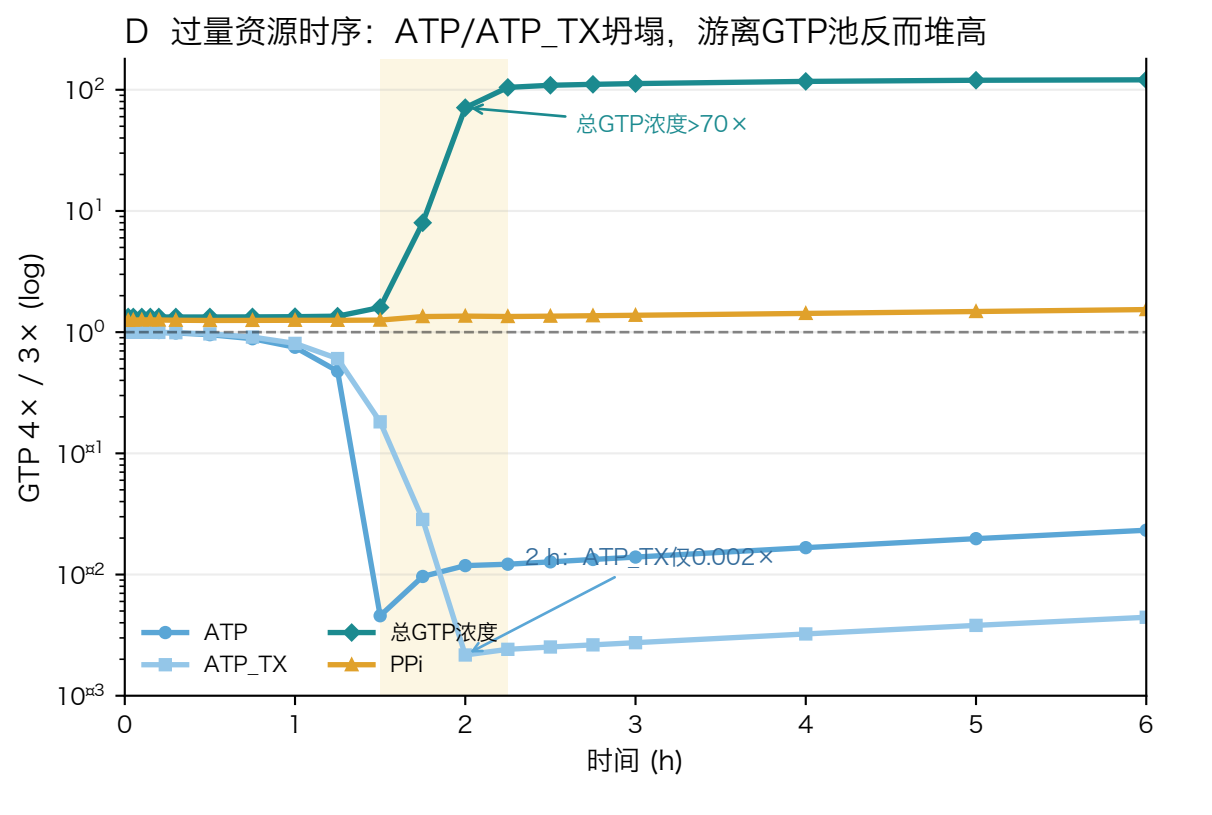
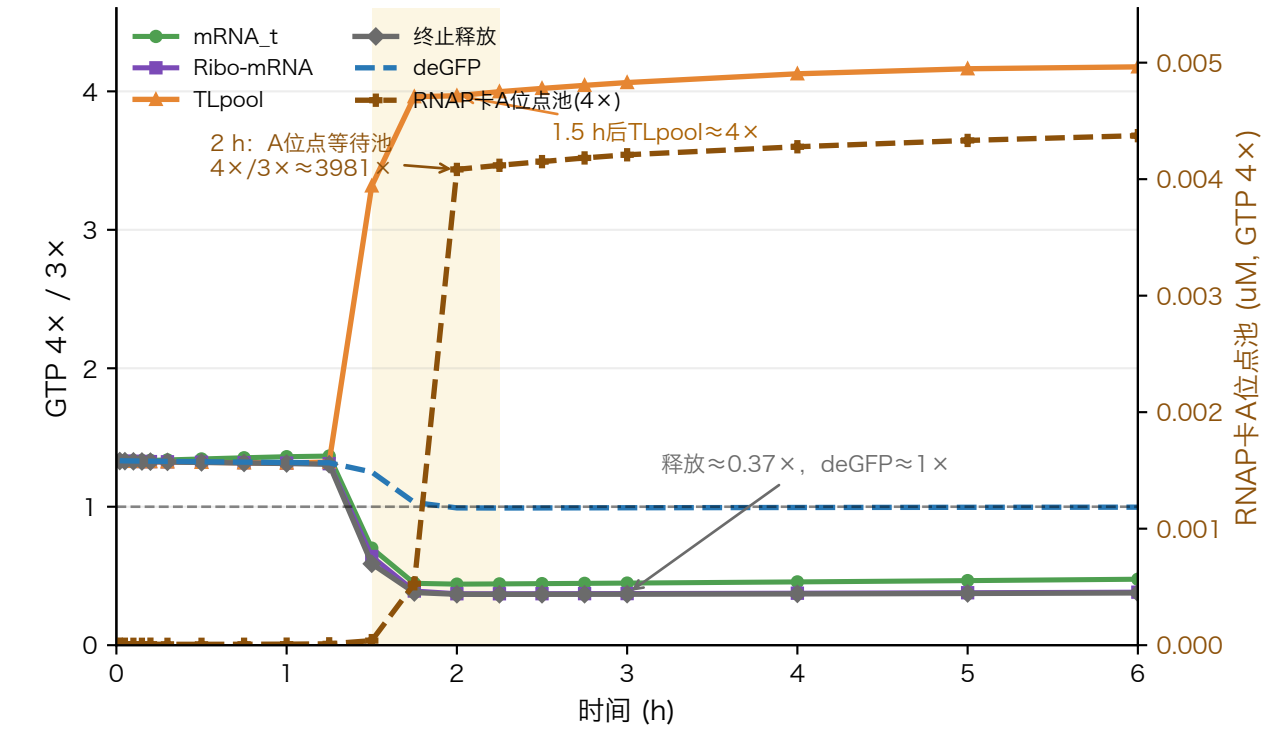
GTP 单独扰动：3×形成高产匹配，4×过量后ATP支撑坍塌并出现翻译链堵塞

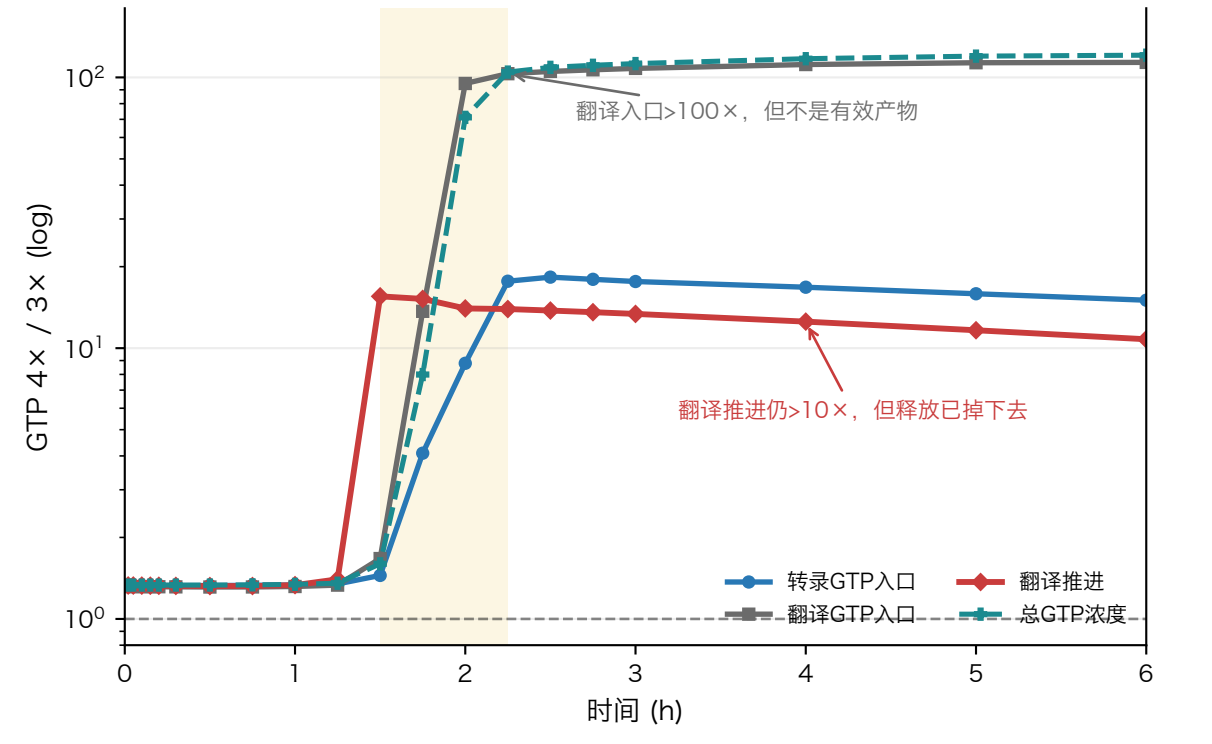












注：D - F 使用统一重算时间点，比较口径为相邻过最优 GTP 4×/3×；总GTP浓度=GTP+GTP_TX。E图RNAP卡A位点池为下一步需要ATP_TX的209个T7RNAP_bound_mRNA状态求和。